A PROJECT

TITLE ON

AI-Powered Chronic Kidney Disease Risk Prediction and Analysis

project report submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING

BACHELOR OF TECHNOLOGY

in

A.Y.: 2026-27

by

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1. Abstract:

Chronic Kidney Disease (CKD) is a serious condition that requires early detection to prevent severe health complications. This project, "CKD Prediction Using XGBoost & LightGBM," uses machine learning techniques to predict CKD based on patient clinical data. The dataset is preprocessed by handling missing values, encoding categorical features, and performing exploratory data analysis. Two gradient boosting algorithms, XGBoost and LightGBM, are trained and compared using performance metrics such as accuracy, precision, recall, and F1-score. The best-performing model is deployed through a Streamlit web application, enabling users to enter patient details and receive instant CKD predictions. This project provides an accurate, efficient, and user-friendly solution that supports early diagnosis and assists healthcare professionals in making informed clinical decisions.

2. INTRODUCTION:

Chronic Kidney Disease (CKD) is a long-term medical condition in which the kidneys gradually lose their ability to filter waste and excess fluids from the blood. If left undetected, CKD can lead to severe complications, including kidney failure and cardiovascular diseases. Early diagnosis plays a vital role in improving treatment outcomes and reducing mortality.With the advancement of Artificial Intelligence and Machine Learning, healthcare systems can predict diseases more accurately using patient medical data. This project, "CKD Prediction Using XGBoost & LightGBM," aims to develop a reliable prediction system using advanced gradient boosting algorithms. The model is trained on clinical data containing patient attributes such as blood pressure, blood glucose, serum creatinine, hemoglobin, and albumin.

The project involves data preprocessing, feature engineering, model training, and performance evaluation to identify the most accurate prediction model. The final model is deployed through a user-friendly interface, enabling healthcare professionals to obtain quick and reliable predictions. This system supports early diagnosis, improves clinical decision-making, and demonstrates the practical application of

machine learning in the healthcare domain.

3. SOFTWARE REQUIREMENTS:

➔ Operating System	-	Windows 10/11, Linux, or macOS
➔ Programming Language	-	Python 3.10 or above
➔ Development Environment	-	Jupyter Notebook / Google Colab / VS Code
➔ Libraries ,	-	Pandas, NumPy, Scikit-learn, XGBoost, LightGBM, Matplotlib, Seaborn, SHAP, Joblib
➔ Web Framework	-	Streamlit
➔ Database	-	SQLite / MySQL
➔ Version Control	-	Git & GitHub
➔ Package Manager	-	pip
➔ Dataset	-	UCI Chronic Kidney Disease Dataset
➔ Browser	-	Google Chrome / Microsoft Edge

Signature of the Guide **Course Instructor**